

## Test Report

Report No.: DGC260616047RE01

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**Applicant** : Shenzhen INNO Technology Co., LTD  
**Address** : F9, Building 20, Xiangnan District 4, Zhangkeng Community, Minzhi Street, Longhua District, Shenzhen, Guangdong, China

**The following sample(s) was/were submitted and identified on behalf of the client as:**

**Product Name** : USB2RS232-C  
**Manufacturer** : Shenzhen INNO Technology Co., LTD  
**Address** : F9, Building 20, Xiangnan District 4, Zhangkeng Community, Minzhi Street, Longhua District, Shenzhen, Guangdong, China

**Date of Sample Received** : Jun. 16, 2026  
**Test period** : Jun. 16, 2026 - Jun. 26, 2026

**Test requested**

In accordance with RoHS Directive 2011/65/EU and amendment 2015/863/EU, to determine Cadmium (Cd), Lead (Pb), Mercury (Hg), Chromium (Cr (VI)), PBBs, PBDEs, Di (2-ethyl hexyl)-phthalate (DEHP), Dibutyl phthalate (DBP), Butylbenzyl phthalate (BBP), Diisobutyl phthalate (DIBP) content on submitted samples.

**Conclusion**

Pass

**Approved by:**

*Richard Ke*  
Richard Ke  
(Signed for and on behalf)

**Date:**

Report Seal

Jun. 26, 2026



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## Test method:

### 1. For the Cadmium (Cd), Lead (Pb),Mercury (Hg), Chromium (Cr (VI)), PBBs, PBDEs:

With reference to IEC 62321 Procedures for the Determination of Levels of Regulated Substances  
in

Electrotechnical Products, XRF scanning first test, then using chemical test method to confirm.

| Testing Item      |                     | Test Method                  | Measuring Instrument | MDL        |
|-------------------|---------------------|------------------------------|----------------------|------------|
| Screening test    |                     | IEC 62321-3-1: 2013 scanning | XRF                  | --         |
| Wet Chemical test | Lead (Pb)           | IEC 62321-5: 2013            | ICP-OES              | 2mg/kg     |
|                   | Cadmium (Cd)        | IEC 62321-5: 2013            | ICP-OES              | 2mg/kg     |
|                   | Mercury (Hg)        | IEC 62321-4: 2013+AMD1:2017  | ICP-OES              | 2mg/kg     |
|                   | Chromium (Cr (VI))▼ | IEC 62321-7-2:2017           | UV-Vis               | 8mg/kg     |
|                   |                     | IEC 62321-7-1: 2015          |                      | 0.10µg/cm² |
|                   | PBBs, PBDEs         | IEC 62321-6: 2015            | GC-MS                | 5 mg/kg    |

### 2. For the DEHP, DBP, BBP and DIBP:

| Testing Item                        | Pretreatment Method | Measuring Instrument | MDL     |
|-------------------------------------|---------------------|----------------------|---------|
| Di (2-ethyl hexyl)-phthalate (DEHP) | IEC 62321-8: 2017   | GC-MS                | 30mg/kg |
| Butylbenzyl phthalate (BBP)         |                     |                      | 30mg/kg |
| Dibutyl phthalate (DBP)             |                     |                      | 30mg/kg |
| Diisobuty phthalate (DIBP)          |                     |                      | 30mg/kg |

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## Photograph of Received Sample :



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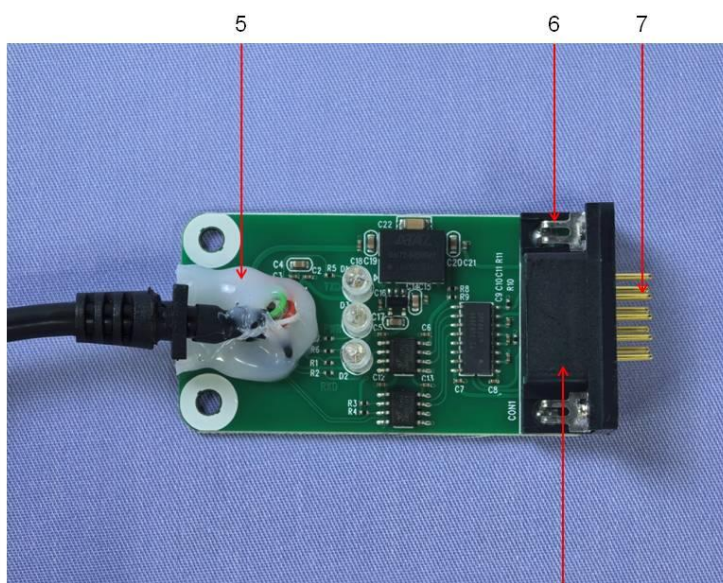
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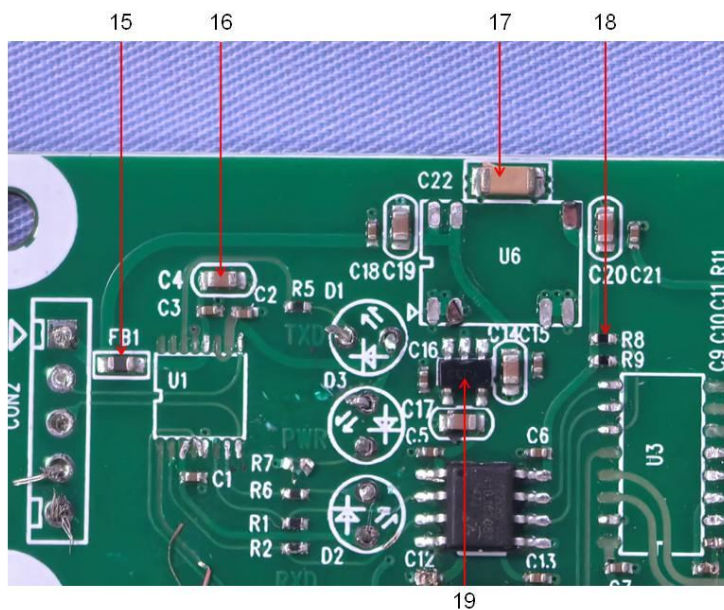
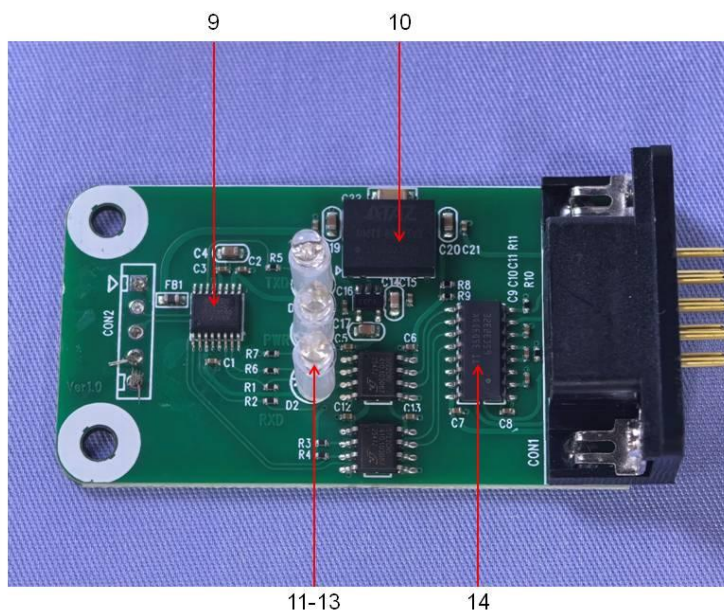
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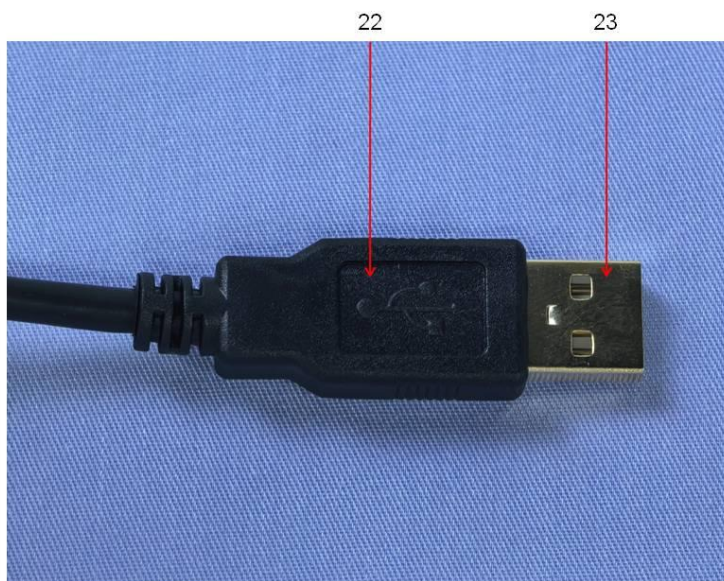
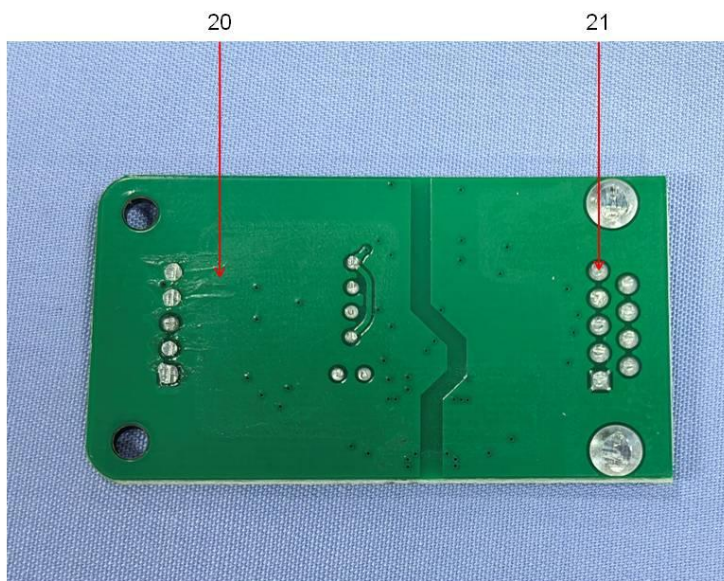
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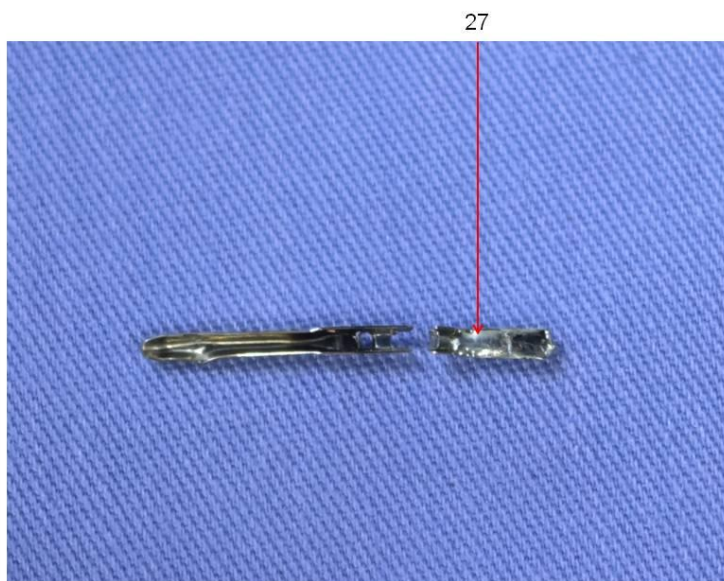
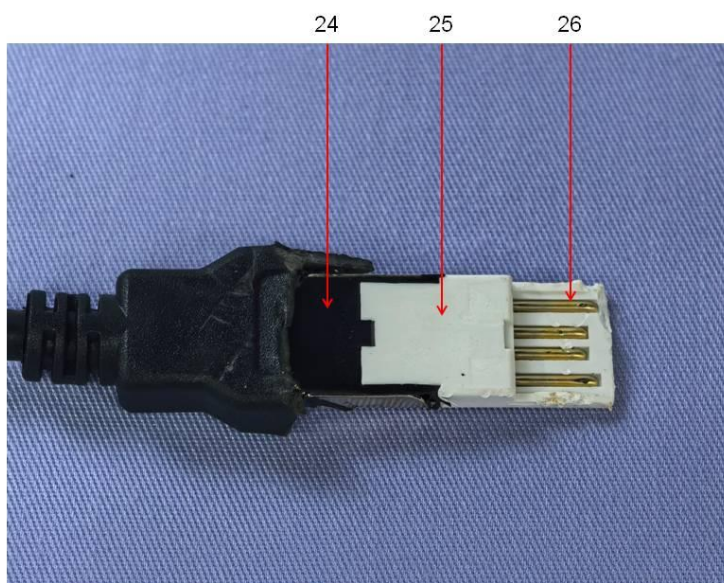




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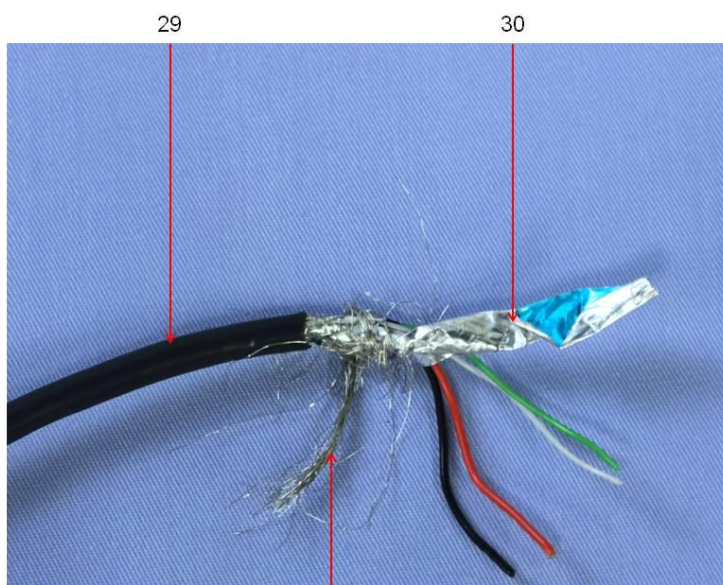
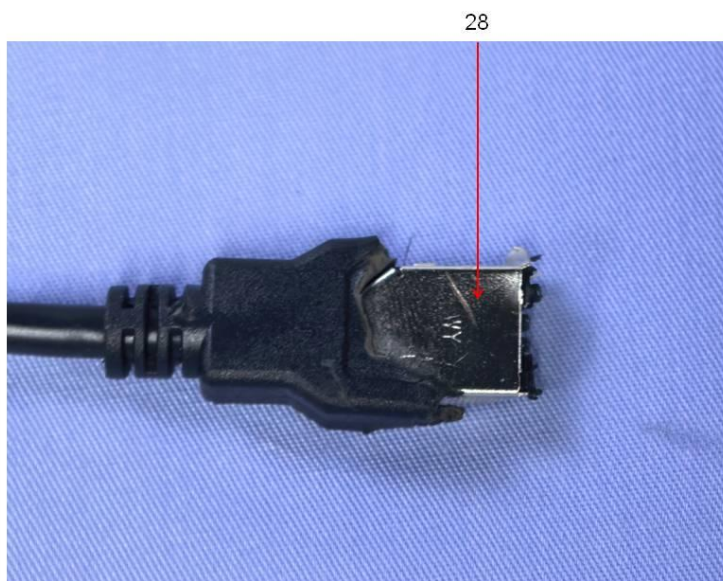
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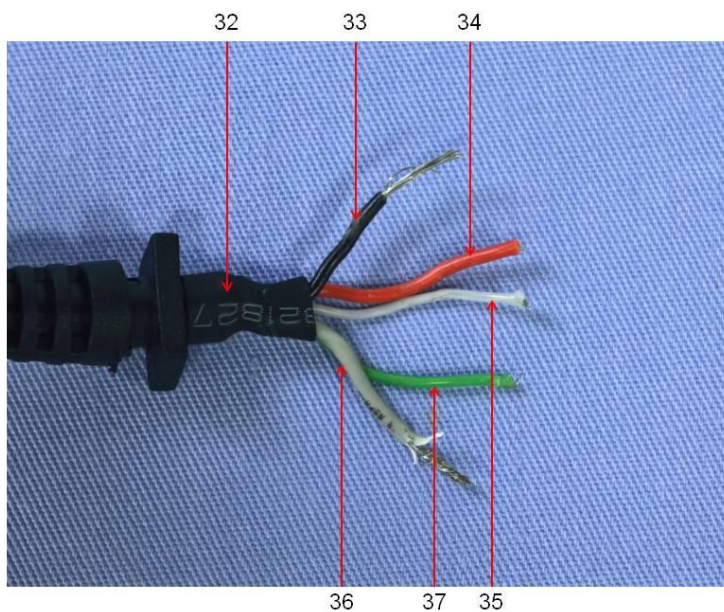




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## 1. Description of the test subject:

| Sample No. | Location           | Sample Description             |
|------------|--------------------|--------------------------------|
| 1          | Shell component    | Green film with black printing |
| 2          | Shell component    | Black plastic shell            |
| 3          | Shell component    | Silver hex screw               |
| 4          | Shell component    | Silver-gold metal frame        |
| 5          | PCB board assembly | White adhesive                 |
| 6          | PCB board assembly | Silver metal sheet             |
| 7          | PCB board assembly | Silver/gold metal pin          |
| 8          | PCB board assembly | Black plastic terminal         |
| 9          | PCB board assembly | SMD IC                         |
| 10         | PCB board assembly | SMD IC                         |
| 11         | PCB board assembly | LED body                       |
| 12         | PCB board assembly | LED lead                       |
| 13         | PCB board assembly | Translucent plastic sleeve     |
| 14         | PCB board assembly | SMD IC                         |
| 15         | PCB board assembly | SMD ferrite bead               |
| 16         | PCB board assembly | Brown SMD capacitor            |
| 17         | PCB board assembly | Light brown SMD capacitor      |
| 18         | PCB board assembly | SMD resistor                   |
| 19         | PCB board assembly | SMD IC                         |
| 20         | PCB board assembly | Green PCB board                |
| 21         | PCB board assembly | Green PCB board solder         |
| 22         | Wire component     | USB plug black plastic grip    |
| 23         | Wire component     | USB plug gold metal frame      |
| 24         | Wire component     | USB plug black soft rubber     |
| 25         | Wire component     | USB plug white plastic         |
| 26         | Wire component     | USB plug silver/gold pins      |
| 27         | Wire component     | USB pin wiring terminal solder |
| 28         | Wire component     | USB plug silver metal frame    |
| 29         | Wire component     | Black wire protector sleeve    |
| 30         | Wire component     | Silver/blue metal foil         |
| 31         | Wire component     | Silver metal wire core         |
| 32         | Wire component     | Black heat shrink tube         |
| 33         | Wire component     | Black wire insulation          |
| 34         | Wire component     | Red wire insulation            |

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|    |                |                            |
|----|----------------|----------------------------|
| 35 | Wire component | Gray-white wire insulation |
| 36 | Wire component | Gray wire insulation       |
| 37 | Wire component | Green wire insulation      |

## 2. Test results (Unit: mg/kg):

| No. | Test Method | Heavy Metals and Flame Retardants |     |     |         |                  | Phthalates |      |      |      | Conclusion |
|-----|-------------|-----------------------------------|-----|-----|---------|------------------|------------|------|------|------|------------|
|     |             | Cd                                | Pb  | Hg  | Cr (VI) | Br (PBBs, PBDEs) | DEHP       | BBP  | DBP  | DIBP |            |
| 1   | Screening   | BL                                | BL  | BL  | BL      | BL               | BL         | BL   | BL   | BL   | Pass       |
| 2   | Screening   | BL                                | BL  | BL  | BL      | BL               | BL         | BL   | BL   | BL   | Pass       |
| 3   | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 4   | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 5   | Screening   | BL                                | BL  | BL  | BL      | BL               | BL         | BL   | BL   | BL   | Pass       |
| 6   | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 7   | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 8   | Screening   | BL                                | BL  | BL  | BL      | IN               | BL         | BL   | BL   | BL   | Pass       |
|     | Wet Chem.   | ---                               | --- | --- | ---     | N.D.             | ---        | ---  | ---  | ---  |            |
| 9   | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 10  | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 11  | Screening   | BL                                | BL  | BL  | BL      | IN               | BL         | BL   | BL   | BL   | Pass       |
|     | Wet Chem.   | ---                               | --- | --- | ---     | N.D.             | ---        | ---  | ---  | ---  |            |
| 12  | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 13  | Screening   | BL                                | BL  | BL  | BL      | BL               | BL         | BL   | BL   | BL   | Pass       |
| 14  | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 15  | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 16  | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 17  | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 18  | Screening   | BL                                | BL  | BL  | IN      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
|     | Wet Chem.   | ---                               | --- | --- | N.D.    | ---              | ---        | ---  | ---  | ---  |            |
| 19  | Screening   | BL                                | BL  | BL  | BL      | BL               | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 20  | Screening   | BL                                | BL  | BL  | BL      | BL               | BL         | BL   | BL   | BL   | Pass       |
| 21  | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 22  | Screening   | BL                                | BL  | BL  | BL      | BL               | BL         | BL   | BL   | BL   | Pass       |
| 23  | Screening   | BL                                | BL  | BL  | BL      | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |

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| No. | Test Method | Heavy Metals and Flame Retardants |    |    |             |                  | Phthalates |      |      |      | Conclusion |
|-----|-------------|-----------------------------------|----|----|-------------|------------------|------------|------|------|------|------------|
|     |             | Cd                                | Pb | Hg | Cr (Cr(VI)) | Br (PBBs, PBDEs) | DEHP       | BBP  | DBP  | DIBP |            |
| 24  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 25  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 26  | Screening   | BL                                | BL | BL | BL          | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 27  | Screening   | BL                                | BL | BL | BL          | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 28  | Screening   | BL                                | BL | BL | BL          | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 29  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 30  | Screening   | BL                                | BL | BL | BL          | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 31  | Screening   | BL                                | BL | BL | BL          | N.A.             | N.A.       | N.A. | N.A. | N.A. | Pass       |
| 32  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 33  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 34  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 35  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 36  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |
| 37  | Screening   | BL                                | BL | BL | BL          | BL               | BL         | BL   | BL   | BL   | Pass       |

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## Note:

- (1) (a) It is the result on total Br while test PBBs, PBDEs by XRF, It is the result on total Cr while test Cr (VI) by XRF.  
 (b) Results are obtained by XRF for primary screening and further chemical testing by ICP-OES (for Pb, Cd and Hg), UV-Vis (for Cr (VI)) and GC-MS (for PBBs, PBDEs) is recommended to be performed, if the concentration exceeds the below warning value according to IEC 62321-3-1:2013 (unit: mg/kg).

| Element       | Polymer                          | Metal                            | Composite Materials              |
|---------------|----------------------------------|----------------------------------|----------------------------------|
| Cadmium (Cd)  | $BL \leq 70 < X < 130 \leq OL$   | $BL \leq 70 < X < 130 \leq OL$   | $LOD < X < 150 \leq OL$          |
| Lead (Pb)     | $BL \leq 700 < X < 1300 \leq OL$ | $BL \leq 700 < X < 1300 \leq OL$ | $BL \leq 500 < X < 1500 \leq OL$ |
| Mercury (Hg)  | $BL \leq 700 < X < 1300 \leq OL$ | $BL \leq 700 < X < 1300 \leq OL$ | $BL \leq 500 < X < 1500 \leq OL$ |
| Chromium (Cr) | $BL \leq 700 < X$                | $BL \leq 700 < X$                | $BL \leq 500 < X$                |
| Bromine (Br)  | $BL \leq 300 < X$                | --                               | $BL \leq 250 < X$                |

- (c) The XRF screening test for RoHS elements –The reading may be different to the actual content in the sample be of non-uniformity composition.
- (d) The Screening results of Phthalates are for primary screening, and further chemical testing by GC-MS are recommended to be performed if the concentration exceeds the warning value. Where n= number of mixed tests.

| Compound | Polymer              |
|----------|----------------------|
| DBP      | $BL \leq 1000/n < X$ |
| BBP      | $BL \leq 1000/n < X$ |
| DEHP     | $BL \leq 1000/n < X$ |
| DIBP     | $BL \leq 1000/n < X$ |

- (e) OL=Over Limit, BL=Below Limit, IN=Inconclusive, LOD= Limit of Detection;  
 (2) mg/kg=ppm=0.0001%, N.D.=Not detected(<MDL), MDL=Method Detection Limit,  
 “---”=Not conducted, “--”=Not regulated, “N.A.”=Not applicable.

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(3)“▼” =Metal sample

a. The sample is positive for Cr (VI) if the Cr (VI) concentration is greater than 0.13  $\mu\text{g}/\text{cm}^2$ .

The sample coating is considered to contain Cr (VI) ;

b. The sample is negative for Cr (VI) if Cr (VI) concentration is less than 0.10  $\mu\text{g}/\text{cm}^2$ .

The coating is considered a non-Cr (VI) based coating ;

c. The result between 0.10  $\mu\text{g}/\text{cm}^2$  and 0.13  $\mu\text{g}/\text{cm}^2$  is considered to be inconclusive

- unavoidable coating variations may influence the determination ;

Information on storage conditions and production date of the tested sample is unavailable and thus Cr (VI) results represent status of the sample at the time of testing.

(4) RoHS Requirement

| Restricted substances                  | Limits          |
|----------------------------------------|-----------------|
| Lead (Pb)                              | 0.1% (1000 ppm) |
| Cadmium (Cd)                           | 0.01% (100 ppm) |
| Chromium(VI) (Cr (VI))                 | 0.1% (1000 ppm) |
| Mercury (Hg)                           | 0.1% (1000 ppm) |
| Polybrominated biphenyls (PBBs)        | 0.1% (1000 ppm) |
| Polybrominated diphenyl ethers (PBDEs) | 0.1% (1000 ppm) |
| Di (2-ethyl hexyl)-phthalate (DEHP)    | 0.1% (1000 ppm) |
| Butylbenzyl phthalate (BBP)            | 0.1% (1000 ppm) |
| Dibutyl phthalate (DBP)                | 0.1% (1000 ppm) |
| Diisobutyl phthalate (DIBP)            | 0.1% (1000 ppm) |

The above limits are reference with RoHS Directive 2011/65/EU and amendment 2015/863/EU.

(5) Unless otherwise specified, refer to ILAC-G8:09/2019 and use the binary decision rule of simple acceptance (W=0) for conformity assessment.



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**声明: Statement:**

1.检测报告无批准人签字、“检测专用章”或“报告章”无效;

1.This report is considered invalid without approved signature, Detection special seal or Report seal;

2.样品及样品信息由申请者提供, 申请者应对其真实性负责, NTEK 未核实其真实性;

2. The sample(s) and sample information was/were provided by the client who should be responsible for the authenticity which NTEK hasn't verified;

3. 本报告检测结果仅对受测样品负责;

3. The result(s) shown in this report refer(s) only to the sample(s) tested;

4. 未经 NTEK 书面同意, 不得部分复制本报告;

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5. In case of any discrepancy between the English version and Chinese version of the testing reports( if generated), the Chinese version shall prevail.

备注:报告未加盖 CMA 资质章时, 本报告中的数据结果仅供科研、教学、企业内部质量控制、企业产品研发等目的用。

Remark: When the report without CMA qualification seal, the testing data and result(s) in this report is(are) just for scientific research, education, internal quality control and product development etc.

\*\*\* End of Report\*\*\*